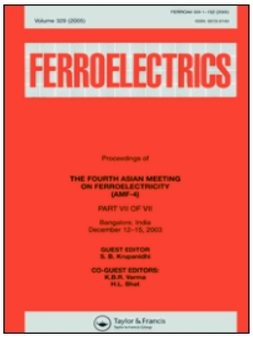

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A brief review on perovskite multiferroics

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ABSTRACT
The paper first introduces the basic physics behind perovskite multiferroics. Then the recent progress on simple, complex perovskites, and manganites are briefly reviewed. The review discloses two important directions on achievement multiferrocity in perovskite: pay with strain engineering and adjust octahedral deformation. The authors also suggest that specific issues like leakage current and characterization platforms are also very important for the study of multiferroics.

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Multiferroic; ferroelectricity; magnetism; perovskite

1. Introduction

Multiferroics originally refers to a single phase material with two or more primary ferroic order parameters, such as ferromagnetism, ferroelectricity and ferroelasticity [1]. The current researches focus on materials exhibiting both magnetism and ferroelectricity simultaneously, which have attracted more attentions for interesting physics and potential applications in the memory devices [2,3], especially after the discoverings of magnetic control polarization in $TbMnO_3$ (TMO) [4] and large polarization in $BiFeO_3$(BFO) [5]. Multiferroics with strong magnetoelectric (ME) coupling at room temperature are the most promising materials for data storage. Unfortunately, TMO has very low ordering temperatures and BFO does not show strong direct ME coupling. Such disadvantages push further more studies, which are clearly manifested by the exponential growth of publications in the area [6].

From a structural view, both TMO and BFO coincidentally belong to perovskite structure, which is perhaps the most adjustable structure model to induce fantastic properties like superconductivity, ferroelectricity, and ferromagnetism. Nevertheless, in general the chance to combine both ferroelectric and ferromagnetic in single perovskite is of scarcity [78]. Therefore, sophisticated strategies have been exploited to combine both magnetic and ferroelectric properties in perovskite-type materials. The goal of the review is to pay attention to the recent progress on single phase perovskite multiferroics, especially emphasizing on the relations between the freedoms of structure and the possibilities of combining magnetization and ferroelectricity in the same phase at room temperature with strong magnetoelectric coupling. The paper is scheduled as followings. First, a brief physical basis of perovskite

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multiferroic will be introduced. Then, the recent progress on achieving multiferroicity in perovskites will be given. Finally, the paper will close with a short summary and perspectives.

2. Physics of perovskite multiferroic materials

2.1. The origins of ferroelectric and magnetic orders in perovskite

The ideal perovskite crystallizes in $Pm\overline{3}m$ space group, which is simple primitive cubic lattice without any deformation, as depicted in fig. 1. The chemical formula of perovskite is commonly expressed as ABO_3, where A and B are two cations of very different chemical valences (2+ and 4+ correspondingly) and sizes, and O is oxygen anion that bonds to both of them, note that both A and B could be mixed cations of different chemical valences. The small cation B^{4+} is surrounded by an octahedron of 6-nearest neighbor O^{2-} (BO_6) and the large A^{2+} is coordinated with maximum 12-nearest neighbor O^{2-}. Such coordination form is stable if the sizes of A^{2+}, B^{4+} and O^{2-} vary under the limitation of famous Goldshmidt tolerance factor t [9]:

$$t = \frac{r_A + r_O}{\sqrt{2}(r_A + r_O)} \tag{1}$$

The t factor somehow decides the lattice types of perovskite [10,11] and the delicate balance could be further influenced by temperature, pressure, electric field, and etc..

The origin of ferroelectricity in perovskite has long been well established by soft mode theory [12]: the lowest-frequency polar-phonon mode becomes softer during cooling, and finally goes to zero frequency at T_C, which is the critical temperature that the material transforms from ideal cubic to lower symmetrical ferroelectric structure. Obviously, such structural instability results in

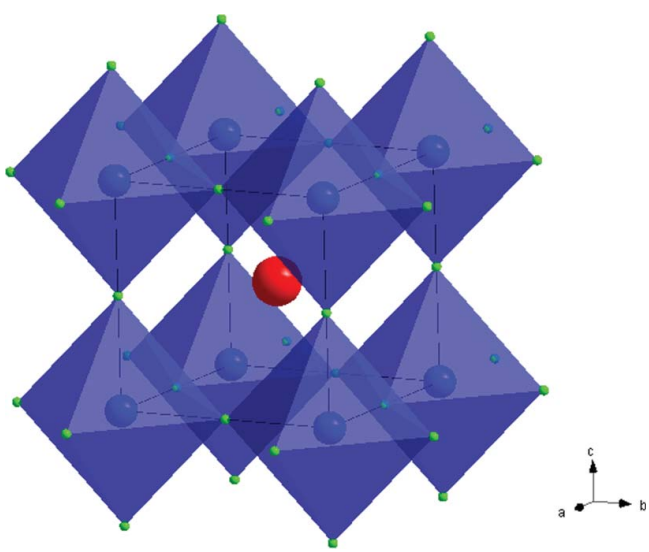

Figure 1. Octahedral framework of ideal cubic perovskite structure, in which the red ball indicates A^{2+} ion, blue balls represents B^{4+} ions, and oxygen ions are represented by green balls. The dashed lines show cubic cell.

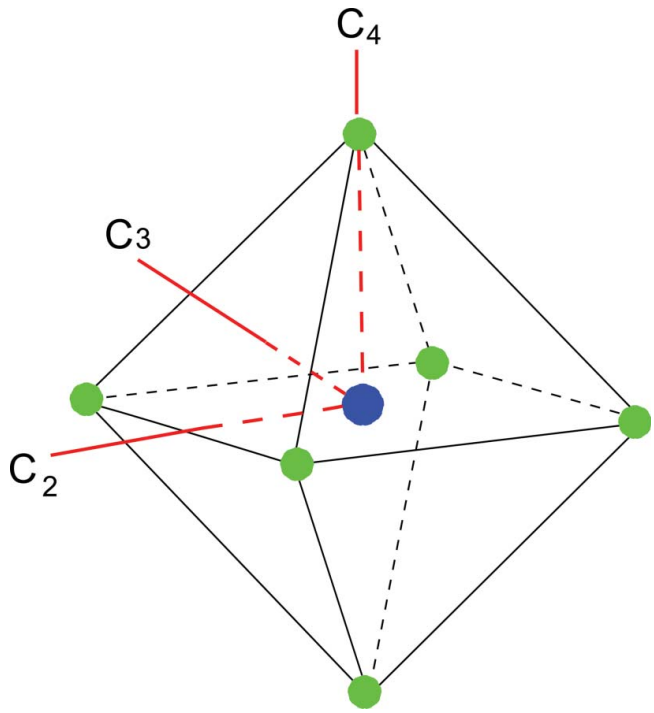

Figure 2. The potential shifts of B-site cation (blue ball) in an octahedral framework of ideal cubic perovskite structure. The oxygen ions are represented by green balls and red lines show high symmetry axes along which B-site cation moves.

broking of centrosymmetry and deformation of perovskite, commonly known as second-order Jahn-Teller Effect [13]. For instance, in classic ferroelectric $BaTiO_3$, Ti^{4+} shifts along C_4, C_2 and C_3 axises (see fig. 2) and stabilizes in different temperature ranges with tetragonal, orthorhombic or rhombohedral structures with different spontaneous polarizations [14]. Most of ferroelectric perovskite oxides, with transition metal ions at B sites as $BaTiO_3$, have empty d electron (d^0). But, keeping in mind, this is not necessary for A-site driven ferroelectrics such as Pb^{2+} and Bi^{3+} based oxides. Moreover, in both B-site driven ferroelectrics and A-site driven ferroelectrics, the force favoring ferroelectric state is the long-range coulomb interaction, shortening one or several B-O or A-O distances [15].

Different to ferroelectricity, the magnetism is governed by quantum-mechanical exchange interactions, including double exchange interaction and superexchange interaction. The double exchange interaction induces ferromagnetism [16,17] which is the most promised magnetic ordering for multiferroic materials used in multi-state memory devices. Unfortunately, the perovskite materials with ferromagnetic ordering commonly bring itinerant electrons, which increase the conductivity and destroy ferroelectricity as they screen the long-range coulomb interactions even if such interactions exist. Thus, the additional attention must be paid to confirm the ferroelectricity in magnetic perovskites [18,19]. In general, the perovskite compounds with double exchange interaction can hardly be multiferroic materials.

Comparison with double interaction, the superexchange interaction in perovskite-type oxides can induce either ferromagnetism or antiferromagnetism [20–24]. Moreover, these compounds are commonly insulators, giving ways to combine ferroelectricity and

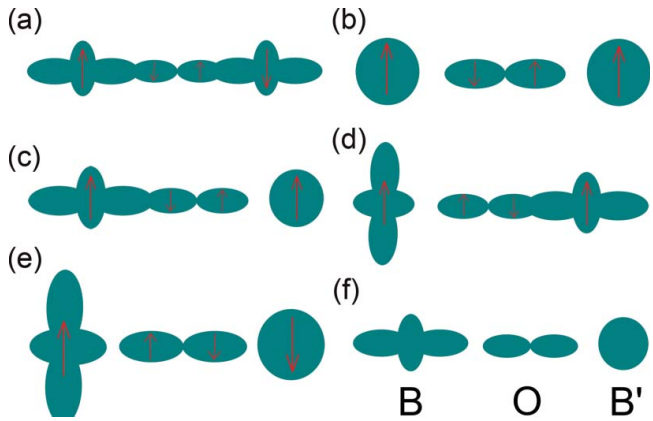

Figure 3. Strong antiferromagnetic order (a), weak antiferromagnetic order (b), weak ferromagnetic order (c), ferromagnetic order (d), and antiferromagnetic order (e) induced by superexhange interaction. (f) represents the legends of electronic orbits for B, B′ and O ions.

magnetism together. For magnetic perovskite oxides, B site cation has partly filled d electrons (d^n) and splits into two energy states in oxygen octahedral environment: two high-energetic e_g orbitals ($d_{x^2-y^2}$, $d_{3z^2-r^2}$) and three low energy t_{2g} orbitals (d_{xy}, d_{yz}, d_{zx}) [25]. The orbitals of two neighboring magnetic cations, coupling through non-magnetic O^{2-}, can be parallel or orthogonal, resulting in different magnetic orders (see fig. 3). In short, it is the configurations of d electrons deciding the magnetic order of materials. For instance, in BFO, $Pb(Fe_{2/3}W_{1/3})O_3$ (PFW), $PbFe_{0.5}Nb_{0.5}O_3$ (PFN) and $PbFe_{0.5}Ta_{0.5}O_3$ (PFT), in which B site occupied by $Fe^{3+}(3d^5, S = 5/2)$ with each d orbital filled by single electron, the orbitals of neighboring Fe^{3+} are parallel with opposite spin directions, producing G-type antiferromagnetism through Fe^{3+}-O^{2-}-Fe^{3+} (see fig. 3(a)). Similarly, ferromagnetism could be produced if the neighboring orbitals of cations are orthogonal with same spin directions, a case for some double perovskite oxides $A(BB')O_3$, ordered B and B' cations with empty and half-filled e_g orbitals (see fig. 3(c)).

2.2. Dzyaloshinsky-Moriya interaction

The model multiferroic BFO, relaxor ferroelectric PFW, PFN and PFT are antiferromagnetic oxides. But antiferromagnetism is not a primary ferroic order parameter, thus it is less favorable for the innovation of memory devices. However, it was reported that a week ferromagnetism could build on antiferromagnetic order through Dzyaloshinsky-Moriya interaction [26,27], which is commonly expressed as:

$$H_{DM} = D_{BB'}\,(S_B \times S_{B'}) \qquad D_{BB'} \propto (\mathrm{x} \times \mathrm{r}_{BB'}) \tag{2}$$

in which $r_{BB'}$ is the vector connecting the site B to the site B' and x is the displacement of the ligand ion (in perovskite oxides, it is O^{2-}) from the BB' line. From a view of geometry, the antiferromagnetism is based on 180° of B-O-B' bond angle and the bending of B-O-B' bonding in distorted perosvkite would give additional magnetic interaction. By decreasing exchange interaction and increasing Dzyaloshinsky-Moriya interaction, week ferromagnetic could be induced [28].

2.3. Magnetoelectric couling and mutual control

For a multiferroic system, the Landau free energy can be wrote as the function of applied electric field E and applied magnetic field H [6,29]:

$$F(E,\ H) = F_0 - P_iE_i - M_iH_i - \frac{1}{2}\varepsilon_0\varepsilon_{ij}E_iE_j - \frac{1}{2}\mu_0\mu_{ij}H_iH_j - \alpha_{ij}E_iH_j - \frac{1}{2}\beta_{ij}E_iH_jH_k - \frac{1}{2}\gamma_{ijk}H_iE_iE_k \tag{3}$$

in which α describes the linear magnetoelectric couping, and β and γ represent bilinear magnetoelectric couping. P_i and M_i are the spontaneous polarization and magnetization in direction i; ε_0 and μ_0 are the permittivity and permeability of free space; and ε_{ij} and μ_{ij} are the relative dielectric permittivity and relative magnetic permeability. The linear magnetoelectric coupling coefficient α can only occur in ferromagnetic or antiferromagnetic point groups. Moreover, from Eq. 3 we can see that the electric field (or magnetic field) can induce magnetization (or polarization) if α exists in a multiferroic system. Brown proposed the upper boundary of linear ME coupling as [30]:

$$\alpha_{ij} \leq \frac{\sqrt{\varepsilon_{ij}\mu_{ij}}}{c_0} \tag{4}$$

here c_0 is the speed of light in vacuum. The formula highlights that large linear ME coupling can be gotten by large electric permittivity and magnetic permeability even without ferroic orders. Nevertheless, large permittivity and permeability are commonly expected for ferroelectric and ferromagnetic materials, although they are not the prerequisites.

It should be noted at here that α can be zero for a multiferroic material because a multiferroics does not necessary to be a magnetoelectric material [29]. And for the applications in data storage, the most interesting properties are the directions of the magnetization or the spontaneous polarization can be controlled by an electric or magnetic field respectively[31]. Commonly such effects are achieved by complex scenarios and the following formulas are used to evaluate the coupling effects:

$$\begin{aligned} \eta &= \frac{\varepsilon(H) - \varepsilon(0)}{\varepsilon(0)} \times 100\% \\ \eta &= \frac{P(H) - P(0)}{P(0)} \times 100\% \\ \eta &= \frac{\chi(E) - \chi(0)}{\chi(0)} \times 100\% \\ \eta &= \frac{M(E) - M(0)}{M(0)} \times 100\% \end{aligned} \tag{5}$$

Probably the model material for controlling polarization by magnetic field is TMO, in which small polarization (0.08 $\mu C/cm^2$) can be rotated by 90° using a magnetic field at low temperatures [4]. The flop of polarization was firstly explained by Mostovoy [32,33], which was argued [34] and replaced by a trilinear magnetoelctric coupling term in the Landau free energy expansion [35]. From a view of structure, the flop of polarization coincides with a

first-order transition to magnetic commensurate phase[36]. The electric control of spin helicity was also revealed in the material by neutron scattering [37].

To fabricate a four-state memory device, 180° switching of ordering parameter is most promising property. According first principles calculations, this is possible in the pervokiste with magnetic cation occupied A-site [38]. In such compound if antiferromagnetism is built though A-B-A superexchange interaction, a week ferromagnetism could be induced and can be switched between 180° symmetry equivalent states with an applied electric field [38]. However, a rather large distortion of lattice has to be induced to shorten A-B for producing superexchange interaction, because in ideal cubic perovskite A-O distance is shorter than A-B distance. The distortion is related to two different BO_6 rotation modes, which are also termed as antiferrodistortive and service as bridges for polar and magnetic orders.

3. Recent progress on the study of multiferroic perovskite

3.1. Beyond d^0-d^n scenario

As we have mentioned in the pervious section, perovskite magnets require d^n electron configuration with n different from zero while classic FE with d^0 electron configuration, what is so called d^0-d^n dilemma after Hill explaining why common ferromagnetic perovskite oxides can hardly to be ferroelectrics [7,39]. There are at least three ways to beyond d^0-d^n dilemma and achieve multiferroicity in perovskite: (1) Let polar and magnetic active ions occupying different Wyckoff positions. For instance, in $BiFeO_3$, ferroelectric active cation Bi^{3+} occupies A site while magnetic cation Fe^{3+} stays at B site of perovskite. And conversely in $EuTiO_3$, magnetic cation Eu^{2+} occupies A site while ferroelectric Ti^{4+} stays at B-site. (2) Prepare double perovskite in which one of sublattice is magnetic and the other is ferroelectric. (3) Prepare magnetic perovskite in which ferroelectricity is strongly related to spiral magnetic order.

3.2. A-site driven ferroelectric multiferroics

It was commonly believed that the off-center shift of B-site cation is the origin of ferroelectricity in perovskite oxides after the finding of ferroelectric $BaTiO_3$. The study by first principles revealed that the lone-pair electrons of A-site cation also can induce ferroelectricity, as in another classic ferroelectric perovskite $PbTiO_3$ [15]. These type ferroelectric materials can be termed as A-site driven ferroelectrics and can include magnetism through B-site magnetic cation replacement, for instance in $BiFeO_3$ [5] and $BiMnO_3$ (BMO) [40].

After finding of large polarization in BFO thin film [5], there are enormous studies focusing on this A-site driven ferroelectric multiferroics. An excellent review paper has been written by Gustau Catalan and J.F. Scott [41]. Recent research has been focused on cycloidal spin modulation [42–44], canted FM order [45–49], antiferroelectric phase [50], and magnetoelectric coupling [51–53].

One of great achievements in the study of multiferroic BFO is the clarification of canted ferromagnetic order. The bulk BFO does not show magnetic properties because of the existence of cycloidal spin modulation, which is a less favorable case for applications in devices. Surprisingly, interested ferromagnetism was found in BFO thin film [5]. The result was of debates [45,46]. On experimental side, it was shown that the Fe^{3+} reduction [47] and size

effect [48] could induce ferromagnetic property in BFO. On theoretical side, the ferromagnetic property in BFO thin film was explained by spin canting according to first principles method [54], which is the result of oxygen octahedra tilting [49]. The canted ferromagnetism was confirmed recently by polarized small-angle neutron scattering method [55]. And moreover, the it was evidenced that epitaxial strain pays a critical role on magnetic properties of BFO, i.e., pseudo-collinear antiferromagnetic ordering [56,57].Recent studies also find interesting spin glass state at low temperatures in BFO [58,59]. Keep in mind that the existence of ferromagnetism of short-range is the base of spin glass phase transition.

There are also several sophisticated studies on the mutual control of polar and magnetic orders recently. On electric field control of magnetic properties, experimentally it was found that magnetic easy plane could be rotated by 71° polarization switching [53]; the spin waves could be controlled by electric field [60]. Theoretically according to first-principles-based simulations, it was shown that magnetic chirality vector could be switched by 180° polarization reorientation [61]; and the magnetism of multiferroic BFO in $Pmc2_1$ phase could be switched by electric field [62]. On magnetic field control of polarization, it was found that a polarization of 0.02 μC^* cm^{-2} in single crystal BFO because of cycloidal spin order can be controlled by magnetic field [51].

The recent studies on BFO also find that the ferroelectric Curie temperature strongly decreases as strain increases, and the cycloidal spin modulation is destroyed by large strain [57]. We can see that epitaxial strain is an additional freedom to tailor both ferroelectricity and magnetism. The findings are very important for applications since thin film materials have to be used for the fabrications of memories or/and spintonics devices.

BMO is another A-site driven multiferroics, which was firstly theoretical predicted by Hill and Rabe [40]. Different to BFO, BMO displays direct ferromagnetic order below 110K[63] or 100 K [64] because of orbital anharmonicity and next-nearest neighbor vibronic interaction [65]. However, the ferroelectricity of BMO is still with controversy [66]. Previously, the polarization in BMO ceramic was measured by ferroelectric hysteresis loop with remanent polarization P_r values around 0.04-0.06$\mu C/cm^2$ [67,68], but which was argued recently [69]. Moreover, theoretical investigation suggests that BMO is impossible to be a ferroelectrics due to symmetry consideration [70–73]. It was also reported that the thin films of BMO have large polarizations [74–76], i.e., P_r of about 9-16 $\mu C/cm^2$ was found in BMO with double $SrTiO_3$ (STO) buffer layers [76]. This could be related to the epitaxial strain in thin films, but according to first principle calculations the ferroelectricity can not be induced by epitaxial strain [77]. From experiments of synchrotron x-ray diffraction and Raman spectroscopy, it was suggested that ferroelectricity could only appear in BMO at pressure range between 37 GPa and 53GPa [78].

3.3. B-site driven ferroelectric multiferroics

Recently, there are also focused studies on $EuTiO_3$ (ETO), which is a potential B-site driven ferroelectric multiferroics. ETO is a G-type antiferromagnet below 5.3 K [79–81], crystallizing in cubic Pmm space group as STO, so called quantum paraelectric antiferromagnet with strain induced ferroelectricity [82,83]. Although the center symmetry structure prohibits the possibility of ferroelectricity in ETO, the STO-like antiferrodistortive phase transition exists in it, which is believed to be strongly associated with antiferroelectricity [84]. The first principles calculation predicted three possible antiferrodistortive phases [85]. An

antiferrodistortive phase transition was found at 282K [86] by specific heat measurement and 235K [87] by synchrotron x-ray powder-diffraction with I4/mcm space group. However, this phase transition was not confirmed by later experiments[81].

In polycrystalline samples, it was predicted that the strong spin-lattice coupling in fact suppresses the polar order in ETO [88]. Fortunately, under large biaxial compressive strain it was predicted that ETO exhibits strong ferromagnetism (spontaneous magnetization, ~7 Bohr magnetons per Eu) and strong ferroelectricity (spontaneous polarization, $\sim 10\,\mu C/cm^2$) simultaneously [89]. Later experiment has confirmed the multiferroic state with biaxial tension [90], and the estimated polarization could be about $29\,\mu C\cdot cm^{-2}$ [91], waiting to be confirmed by direct ferroelectric hysteresis loop. An additional way to induce ferroelectricity in ETO is doping, i.e., both $Eu_{0.8}Y_{0.2}MnO_3$ [92] and $Eu_xSr_{1-x}MnO_3$ [93,94] show ferroelectricity.

The strong magnetoelectric coupling in ETO has been confirmed by variations of dielectric (magnetic) properties under the magnetic (electric) field [95–100]. Recently, it was evidenced that electric field can switch the magnetic ground state to ferromagnetic in compression film [101], and such property even exists without strain according to the prediction of Landau-Ginzburg-Devonshire theory [102].

3.4. Double and complex perovskite

Substitution of B with B' in perovskite ABO_3 leads to solid solution $AB_{1-x}B_xO_3$, which is called complex perovskite, or double perovskite if $x = 0.5$ [103]. In fact, before the declaration of Hill on the d^0-d^n dilemma, fabrication of a complex (or double) perovskite, in which one of the sublattices is magnetic and the other consists of ferroelectric d° cation, was historically considered to achieve magnetically ordered ferroelectrics [104] and is still very active in the recent renaissance of multiferroics [105].

The double perovskites multiferroics include Bi_2NiMnO_6 [106], Bi_2FeCrO_6 [107] Lu_2MnCoO_6 [108], Y_2NiMnO_6 [109] and etc. All these compounds have rock-salt-type arrangement of B-site cations in a double-perovskite unit cell. Both polar and magnetic phase transition temperatures are very low for these materials, except Bi_2FeCrO_6 which is a room temperature multiferroics in thin film form [107]

In contrast to double perovskites, $Pb(Fe_{2/3}W_{1/3})O_3$ (PFW), $PbFe_{0.5}Nb_{0.5}O_3$ (PFN) and $PbFe_{0.5}Ta_{0.5}O_3$ (PFT), so called relaxor ferroelectrics with G-type antiferromagnetic order, have relatively high polar and magnetic ordering temperatures, i.e., T_C at 150K (our recent experimental result), 370K [110,111], and 240K [112] respectively, and T_N at 350K [113], 143K [114], and 180K [112] respectively. Among these, PFW, which has the highest magnetic ordering temperature, is very promising for achieving room temperature multiferroicity by preparing solid solutions with classic ferroelectrics, i.e., $PbTiO_3$(PTO) [115–119].

We summarized the results based on experiments of differential scanning calorimetry (DSC), Raman spectra and dielectric measurement of both ceramics and single crystals, and plotted the polar and magnetic phase diagrams in fig. 4. As shown in fig. 4 (a), the ferroelectricity of the solid solution generally enhances with the increasing of PTO content, realizing a relaxor to normal ferroelectic phase transition with structure transiting from pseudo cubic to a P4mm structure. T_c approaches room temperature with 20% (mol) doping of PTO.

In contrast to polar phase diagram, the magnetic phase diagram is of controversy as shown in fig. 4 (b). Two magnetic phase transition temperatures, paramagnetic-antiferromagnetic phase transition temperature T_{N2} and ferromagnetic or spin glass

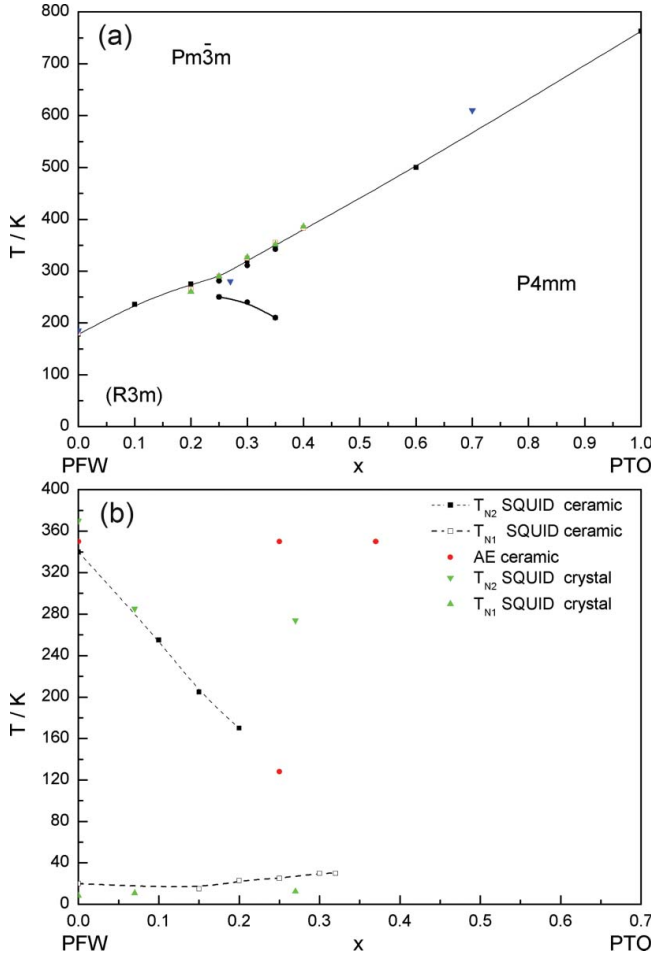

Figure 4. Polar and magnetic phase diagrams of PFW-PTO. For polar phase diagram, except down-triangles which are the results from single crystal data [117], all others are based on experimental date of ceramics based on DSC [115], Raman spectra [118] and dielectric measurement [124]. For magnetic phase diagram, the black squares and red circles are based on SQUID [116] and AE [119] experiments of ceramic, and green triangles are based on SQUID of single crystal [117].

phase transition temperature T_{N1} have been found. With increasing of PTO, SQUID measurement suggests that T_{N2} deceases sharply for ceramic [116] but keeps constant for single crystal [117]. Acoustic emission (AE) measurement indicates it keeps constant even for ceramic [119]. The SQUID measurement suggests spin glass phase at low temperature for single crystal [117].

Recently, there are several results suggesting interesting weak ferromagnetism at room temperature for several PFW-PZO systems, i.e., 0.8PFW-0.2PTO thin films [120], (1-x) PFW- x PTO ceramics (x = 0.10-0.30) [121,122], and PFW itself [121,123]. These findings could be related to occurrence of Fe^{2+} and oxygen vacancies [121], or a weak superexhange of Fe^{3+}-O-B-O-Fe^{3+} in the ordering region [113] as in PFW, or canted antiferromagnetic interaction [54] as in pure BFO. In each of these models, the sub-nano region ordering is critical to induce the attractive magnetic behaviors.

In solid solution systems, more remarkable findings have been found in solid solutions with classic piezoelectric material $PbZr_{0.53}Ti_{0.47}O_3$ (PZT). In PFW-PZT, magnetic control of large polarization at room-temperature has been found [124]. Although there were some controversies [125,126], similar results were found in PFT-PZT [127–130] and PFN-PZT [131]. These compounds with remarkable magnetic, polar and magnetoelectric properties contain only 20% (mole) of Fe at B site randomly distributed. The ferromagnetic and magnetoelectric properties due to spins clustering [127–133], is of short range chemical order. Therefore, ferromagnetic and magnetoelectric properties can exit at room temperature even though the long-range magnetic order establishes far below the room temperature [134].

3.5. Manganites

Manganites are a group of materials exhibiting very interesting magnetoelectric properties, i.e., magnetic control polarization in TMO [4] and electric field control magnetism in $HoMnO_3$ (HMO) [135]. In TMO thin film, a larger temperature range for magnetoelectric coupling has been also reported recently [136]. However, the polarization in these compounds is usually very small. According to first principle calculations, it was predicted that E-type antiferromagnets could have large polarization and magneoelectric coupling, i.e., $6\mu C^{*}cm^{-2}$ in orthorhombic perovskite HMO [137]. The compound has been synthesized by soft-chemistry conditions [138] with large magnetodielectric effects [139], but the measured polarization is still very small, among $0.009\mu C^{*}cm^{-2}$ [[140] and $0.24\mu C^{*}cm^{-2}$ [[141].

$CaMnO_3$ is another manganite in which the ferroelectricity and magnetism is expected to be driven by Mn cation [142]. Although no experimental results have been found, interestingly, attractive multiferroic properties were found in quadruple manganite $CaMn_7O_{12}$ which has fully ordered A-site $Ca^{2+}Mn^{3+}$[[143]. Moreover, because of combination of rotation and tilt of BO_6, strained $Ca_3Mn_2O_7$ should reverse the direction of magnetization by external electric field according to first principle calculations [144,145]. In fact, BO_6 deformation on multiferrocity has been predicted before [146]. More detailed review on perovskite manganites could be found in a recent literature [147].

4. Summary and perspectives

In this brief review, we introduced some recent progress on perovskite muliferroics combining experimental and theoretical investigations. From these fantastic findings it can be seen that strain engineering is a powerful tool to achieve multiferrocity. Materials in thin film form are necessary not for miniaturized microelectronic devices but for unique properties only existing in thin films, i.e., ferroelectricity, magnetism and magnetoelectric coupling.

The findings also emphasize the important of BO_6 deformation on multiferroicity. In G-type antiferromagnetic perovskites,ferromagnetism origins from BO_6 rotation. In manganites, both ferroelectric and magnetic properties could be related to the combination of BO_6 rotation and tilt. In a general consideration, octahedral tilt is one of structural instabilities for both polar and magnetic perovskites. The tilt of BO_6 influences the symmetry of crystal lattice which decides the possibility of ferroelectricity. Moreover, B-O-B bonding is highly controlled by BO_6 tilt. Therefore, both ferroelectric and magnetic properties can be adjusted by the deformations of BO_6.

For perspectives, one big issue for multiferroics is leakage current. The ferroelectricity could be screened by poor insulating property of materials. Therefore, any effects correlated to increasing conductivity should be avoided. For most of perovskite oxides, insulating properties generally deteriorate because of the reduction of magnetic ions and/or oxygen vacancies. Moreover, the bandgap of some multiferroics could be narrower than classic ferroelectrics [148]. As a result, the conductivity generally increases. Thus, except experimental methods to decrease defects, a comprehensive understanding the relation between electronic energy-band structure and conductivity is needed for perovskite multiferroics especially those with narrow bandgaps.

Another deserved perspective for multiferroics is developing new methods and facilities to characterize properties. We can see that several novel multiferroic materials have rather small polarization. For instance, in multiferroic manganites, the typical polarization is 1000 times less than classic ferroelectrics, which gives difficulty to precisely characterize. Alternatively, the spontaneous polarization is commonly calculated by integrating pyroelectric current, a less persuasive way because of possible parasitic effect by trapping carries. Moreover, only few laboratories in the world can perform well established magnetoelectric coupling measurement, which greatly limits the research in the area. Thus, developing simple and standard method to characterization multiferrocity is also of emergence.

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